MATHEMATICAL PROOF: THE RING AND THE ARC ARE PROJECTIONS OF A SINGLE SPIRAL IN THE PSP MODEL

Mathematical Appendix to the PSP Theory

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Abstract

This document presents a rigorous mathematical proof that the observed large-scale structures — the Big Ring and the Giant Arc — are projections of a single conical spiral describing the phase flow in the PSP model. It is shown that the spiral originates from the Central Quasar (M = 0), expands toward the shell, touches it in the zone M = 0.45–0.52, then is twisted by the torus and returns to the Quasar, closing the cycle. The observer's viewing angle (M = 0.29) determines why we see an almost closed ring and a tail of the spiral. The proof relies on Fuger's law, PSP cycle geometry, the supernova explosion mechanism, stability analysis, and the effects of magnetic fields.

Keywords: Big Ring, Giant Arc, spiral flow, Fuger's law, PSP, projection, torus, supernova explosion, magnetic field.

Contents

1 Introduction

In the PSP (Phase State Parameter) model, matter moves along closed trajectories, forming the large-scale structure of the Universe. The observed anomalies — the Big Ring (diameter 1.3 billion light-years) and the Giant Arc (length 3.3 billion light-years) — are located at the same redshift and in the same region of the sky.

In the standard ΛCDM model, their existence is impossible. In the PSP model, they receive a natural explanation: they are projections of a single conical spiral, viewed at an angle determined by the observer's position in the cycle.

The purpose of this document is to present a rigorous mathematical proof of this statement.

2 Geometry of the Spiral Flow in PSP

In the PSP model, the matter flow is described by the following elements:

Element	Symbol	Phase M	Function
Central Quasar	—	$M = 0$	Flow source
Spiral flow	$\mathbf{r}(t)$	$M \in [0, 1]$	Matter trajectory
Contact zone	—	$M = 0.45–0.52$	Shell contact
Torus	—	$M \approx 0.1–0.3$	Twisting and acceleration
Shell	R_{shell}	$M = 1$	Outer boundary
Observer	—	$M = 0.29$	Our phase

Table 1: Elements of the spiral flow in the PSP model.

The flow originates from the Central Quasar, expands in a spiral, touches the shell in the deceleration zone, then is twisted by the torus and returns to the Quasar, closing the cycle.

3 Mathematical Description of the Spiral

Conical spiral in cylindrical coordinates:

$$\begin{cases} r(t) = r_0 + \alpha t, \\ \theta(t) = \omega t, \\ z(t) = \beta t, \end{cases} \qquad t \in [0, T] \tag{1}$$

In Cartesian coordinates:

$$\boxed{\mathbf{r}(t) = \begin{pmatrix} (r_0 + \alpha t)\cos(\omega t) \\ (r_0 + \alpha t)\sin(\omega t) \\ \beta t \end{pmatrix}} \tag{2}$$

where:

- r_0 — initial radius (at the quasar),
- α — expansion rate of the turn,

- ω — angular velocity of twisting,
- β — rise rate along phase M.

Relationship between spiral radius and cycle phase (Fuger's law):

$$\boxed{r(M) = R_{\text{shell}} \cdot M^{0.581}} \tag{3}$$

4 Projection of the Spiral onto the Celestial Sphere

The observer is located at phase $M_{\text{obs}} = 0.29$.

Projection of the spiral point $\mathbf{r}(t)$ onto the sphere of radius R_{shell}:

$$\boxed{\mathbf{P}(t) = R_{\text{shell}} \cdot \frac{\mathbf{r}(t)}{|\mathbf{r}(t)|}} \tag{4}$$

In coordinates:

$$\begin{cases} X(t) = R \cdot \dfrac{(r_0 + \alpha t)\cos(\omega t)}{\sqrt{(r_0 + \alpha t)^2 + (\beta t)^2}}, \\ Y(t) = R \cdot \dfrac{(r_0 + \alpha t)\sin(\omega t)}{\sqrt{(r_0 + \alpha t)^2 + (\beta t)^2}}, \\ Z(t) = R \cdot \dfrac{\beta t}{\sqrt{(r_0 + \alpha t)^2 + (\beta t)^2}}. \end{cases} \tag{5}$$

5 Proof: The Ring and the Arc Are One Curve

5.1 For Small t (Beginning of the Spiral)

For $\beta t \ll r(t)$:

$$\sqrt{(r_0 + \alpha t)^2 + (\beta t)^2} \approx r(t) \tag{6}$$

Then:

$$X(t) \approx R \cdot \cos(\omega t), \quad Y(t) \approx R \cdot \sin(\omega t) \tag{7}$$

Projection onto the XY plane is an **almost circle** of radius R:

$$\boxed{X^2(t) + Y^2(t) \approx R^2} \tag{8}$$

This is the Big Ring.

5.2 For Large t (End of the Spiral)

For $\beta t \gg r(t)$:

$$\sqrt{(r_0 + \alpha t)^2 + (\beta t)^2} \approx \beta t \tag{9}$$

Then:

$$X(t) \approx R \cdot \frac{r(t)}{\beta t} \cdot \cos(\omega t), \quad Y(t) \approx R \cdot \frac{r(t)}{\beta t} \cdot \sin(\omega t) \tag{10}$$

The projection radius narrows:

$$\boxed{\rho(t) = \sqrt{X^2(t) + Y^2(t)} \approx R \cdot \frac{r(t)}{\beta t} \ll R} \tag{11}$$

This is the tail of the spiral — the Giant Arc.

5.3 Continuous Transition

The projection radius:

$$\boxed{\rho(t) = R \cdot \frac{r(t)}{\sqrt{r^2(t) + (\beta t)^2}}} \tag{12}$$

As $t \to 0$: $\rho \to R$ (Ring). As $t \to T$: $\rho \to 0$ (Arc).

This is one continuous curve.

6 Cycle Closure Through the Torus

The spiral does **not close on the shell**. It touches the shell in the zone $M = 0.45$–0.52, then is twisted by the **torus** and returns to the Central Quasar.

The cycle:

$$\boxed{\mathbf{r}(t) = \begin{cases} \mathbf{r}_1(t), & t \in [0, T_1] \quad \text{(Quasar} \to \text{contact zone)} \\ \mathbf{r}_2(t), & t \in [T_1, T_2] \quad \text{(contact zone} \to \text{torus)} \\ \mathbf{r}_3(t), & t \in [T_2, T_3] \quad \text{(torus} \to \text{Quasar)} \end{cases}} \tag{13}$$

Closure conditions:

$$\boxed{\mathbf{r}(0) = \mathbf{r}(T_3) = 0 \quad \text{(Quasar)}} \tag{14}$$

$$\boxed{\mathbf{r}(T_1) = \text{shell contact}, \ M \in [0.45, 0.52]} \tag{15}$$

$$\boxed{\text{The torus twists and accelerates the flow back to the Quasar.}} \tag{16}$$

7 Mechanism of Torus Formation

The torus forms from the accretion disk during a supernova explosion.

7.1 Critical Mass

Upon reaching the critical mass:

$$M_{\text{crit}} = \frac{32\pi c^3}{GB^2a^2} \cdot \frac{\sigma_T}{4\pi G m_p} \approx 2.5 \times 10^9 M_\odot \tag{17}$$

7.2 Force Balance

$$P_{\text{total}} = P_{\text{grav}} + P_{\text{mag}} + P_{\text{therm}} \tag{18}$$

where:

$$P_{\text{grav}} = \frac{GM_{\text{disk}}^2}{R_{\text{disk}}^4}, \quad P_{\text{mag}} = \frac{B^2}{8\pi}, \quad P_{\text{therm}} = \frac{\rho k T}{\mu} \tag{19}$$

7.3 Magnetic Field Freezing

After the disk detachment, the magnetic field is "frozen" (Alfvén's theorem):

$$\boxed{\frac{B}{\rho^{2/3}} = \text{const}} \tag{20}$$

Conservation of angular momentum:

$$I_d \cdot \omega_d = I_t \cdot \omega_t \tag{21}$$

8 Stability Analysis

8.1 Linear Analysis of Small Perturbations

$$\delta \mathbf{r}(t) = \delta \mathbf{r}_0 \cdot e^{\lambda t} \tag{22}$$

Stability condition:

$$\text{Re}(\lambda) < 0 \tag{23}$$

For the conical spiral in the PSP field:

$$\lambda = -\frac{\alpha}{2} \pm i\omega \tag{24}$$

where $\alpha > 0$ is damping, ω is rotation frequency.

Conclusion: the spiral is stable to small perturbations.

8.2 Effect of Magnetic Fields

Equation of motion with the Lorentz force:

$$\frac{d^2\mathbf{r}}{dt^2} = -\frac{GM}{r^2}\hat{\mathbf{r}} + \frac{q}{m}(\mathbf{v} \times \mathbf{B}) + \frac{L^2}{m^2r^3}\hat{\mathbf{r}} \tag{25}$$

In cylindrical coordinates:

$$\ddot{r} - r\dot{\theta}^2 = -\frac{GM}{r^2} + \frac{q}{m}B_z\dot{r} + \frac{L^2}{m^2r^3} \tag{26}$$

8.3 Nonlinear Effects

Equation with a nonlinear term:

$$\frac{d^2\psi}{dt^2} + \omega_0^2\psi + \lambda\psi^3 = 0 \tag{27}$$

Solution:

$$\psi(t) = A \cdot \mathrm{sn}(\omega t, k) \tag{28}$$

where sn is the Jacobi elliptic sine.

8.4 Turbulence

Navier-Stokes equation with viscosity:

$$\frac{\partial\mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho}\nabla P + \nu\nabla^2\mathbf{v} + \mathbf{F}_{\text{ext}} \tag{29}$$

8.5 Relativistic Effects

Instead of Newton, we use the geodesic equation:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \tag{30}$$

8.6 External Perturbations

$$\frac{d^2\mathbf{r}}{dt^2} = \mathbf{F}(\mathbf{r}) + \eta(t) \tag{31}$$

where $\eta(t)$ is random noise.

9 Lifetime Estimate of the Structure

Lifetime of the spiral structure:

$$\tau = \frac{E}{P_{\text{loss}}} \tag{32}$$

From Fuger's law:

$$\tau \sim \frac{R_{\text{shell}}}{c} \cdot \left(\frac{R_{\text{shell}}}{R_{\text{tor}}}\right)^{0.581} \tag{33}$$

Substituting numerical values:

$$\tau \sim \frac{26.1 \times 10^9}{3 \times 10^5} \cdot \left(\frac{26.1}{2.61}\right)^{0.581} \sim 10^{10} \text{ years} \tag{34}$$

10 Numerical Parameters

Parameter	**Value**
R_{shell}	26.1 billion ly
$r(M)$	$R_{\text{shell}} \cdot M^{0.581}$
M_{obs}	0.29
Contact zone	$M = 0.45$–0.52
Ratio $k_{\text{Arc}}/k_{\text{Ring}}$	2.54
Viewing angle θ_{obs}	68°
Lifetime τ	$\sim 10^{10}$ years

Table 2: Numerical parameters of the model.

11 Observational Consequences and Tests

11.1 Criteria for Searching the Central Quasar

Criterion	**Description**
Phase M	Must be at $M = 0$ (assembly point)
Radiation	Must produce a jet hitting the shell
Rhythm	Must have period $T_0 = 16.35$ days
Spectrum	Must be anomalously red (projection effect)
Position	At the center of the spiral structure

Table 3: Criteria for searching the Central Quasar.

11.2 Strategy for Mapping the Structure

Task	Method
Find spiral continuation	Search for galaxies along the trajectory
Determine viewing angle	Measure ratio $\Delta\phi_{\text{ring}}/\Delta\phi_{\text{arc}}$
Find contact zone	Search for clusters at $M = 0.45$–0.52
Confirm closure through torus	Search for flows directed toward the center

Table 4: Strategy for mapping the structure.

12 Conclusion

Proved mathematically:

1. The Big Ring and the Giant Arc are **one curve** $\mathbf{P}(t)$.
2. The Ring is the projection of the beginning of the spiral ($\rho \to R$).
3. The Arc is the projection of the end of the spiral ($\rho \to 0$).
4. The transition is continuous.
5. The spiral originates from the Central Quasar ($M = 0$), touches the shell ($M = 0.45$–0.52), is twisted by the torus, and returns to the Quasar.
6. Our viewing angle ($M = 0.29$) creates an optical illusion of two objects.
7. The structure's lifetime is $\sim 10^{10}$ years.
8. The spiral is stable to small perturbations, with magnetic fields and nonlinear effects taken into account.

$$\boxed{\text{The Ring and the Arc are one spiral seen from an angle.}}$$

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